Sampling variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Confounders and controls. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of sampling variability. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **sampling variability**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **sampling variability**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Confounders and controls. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **sampling variability** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Sampling variability case. Separate observations, given facts and assumptions before choosing an explanation.

Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Sampling variability. Connect each labelled input, deciding step and result.

Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 **without replacement**. Their means are1, 2, 2, 2, 2, 3.

Sampling variability — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.	Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 without replacement.	Mean= (1+2+2+2+2+3)/6=2; mean2 occurs4/6=2/3. The population mean is2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Sampling variability. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

Predicted result

Prediction: The counts should agree unless some condition, data or scoring changed.

Why the result follows

Reason: The sampling explanation no longer applies when the records are identical.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Sampling variability to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Sampling variability	Changes in estimates across eligible samples
Sampling distribution	The distribution of an estimate under a sampling procedure
Without replacement	Selecting distinct units within a sample

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Sampling variability, predict a step, then compare with its explanation.

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.

Mean=(1+2+2+2+2+3)/6=2; mean2 occurs4/6=2/3. The population mean is2, while individual samples differ.

Worked explanation · Sampling variability

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Sampling variability

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')
    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

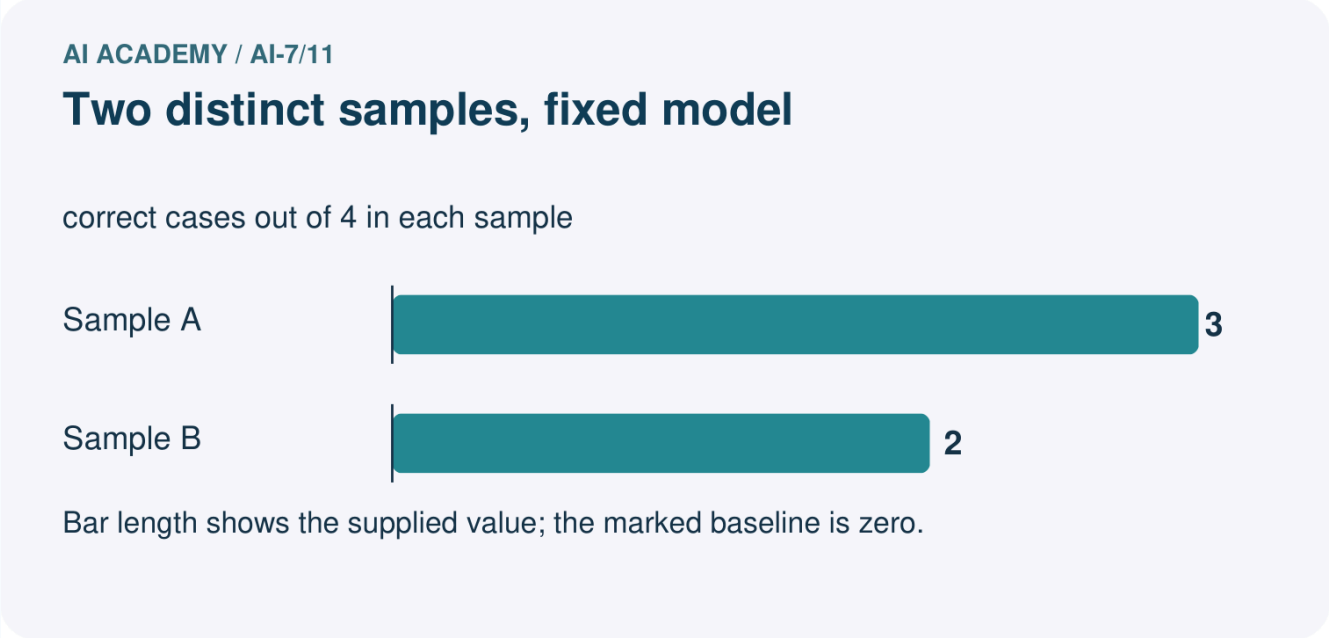
A pictorial worked case: Sampling variability

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

The facts we will use

Supplied facts

Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/11

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep model fixed

No training or threshold change occurs between samples.



2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.



3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep model fixed	No training or threshold change occurs between samples.
2. Compute observed rates	A is 75%; B is 50%, differing by 25 percentage points.
3. Consider sampling	Different included cases can change a small-sample rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

What this example does not establish

Keep the stated limit

These two samples do not supply a probability model or confidence interval; none is invented.

Keep the stated limit

Read the labelled picture: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

Keep the stated limit

Change one thing: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sampling variability claim and explain a repair using the supplied evidence.

This finite example is not a general normality guarantee or a model of a biased convenience sample.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Sampling variability with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Sampling variability**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Sampling variability**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

> **My explanation using a pictured step and the given facts**
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This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Sampling variability. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')
    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sampling variability. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **sampling variability evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sampling variability**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sampling variability case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Sampling variability — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.	Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 without replacement.	Mean= (1+2+2+2+2+3)/6=2; mean2 occurs4/6=2/3. The population mean is2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Sampling variability, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

By the end, I can	How I will show it
Explain the mechanism and terminology of sampling variability	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sampling variability without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **sampling variability**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Different included cases can change a small-sample rate.

1. Consider sampling

2. Keep model fixed

3. Compute observed rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Consider sampling Different included cases can change a small-sample rate. **Sampling variability** means a statistic can vary because different units were observed.

Do not automatically attribute every rate change to a model improvement or deterioration.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Which conclusion is supported by the supplied facts?

1. The counts should agree unless some condition, data or scoring changed.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Repair a missing building block

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

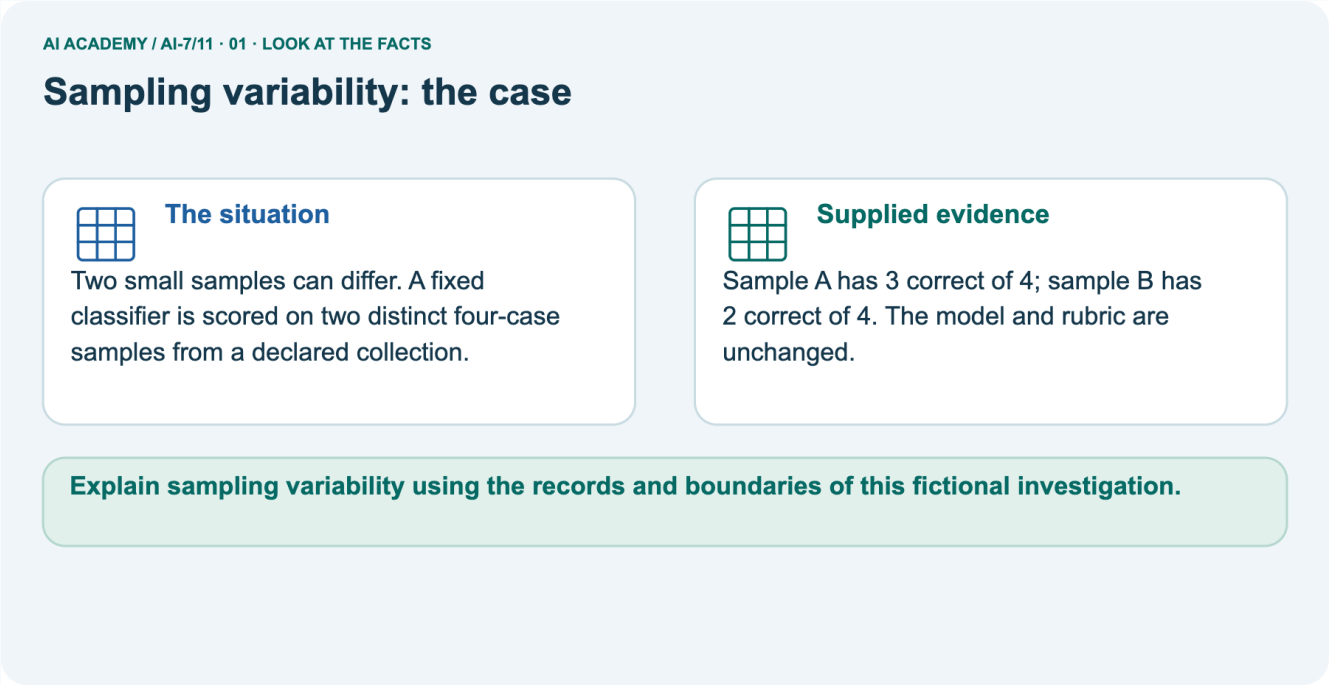
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sampling variability graphic using its labels, deciding evidence and relevant limitation.

01. Sampling variability: the case



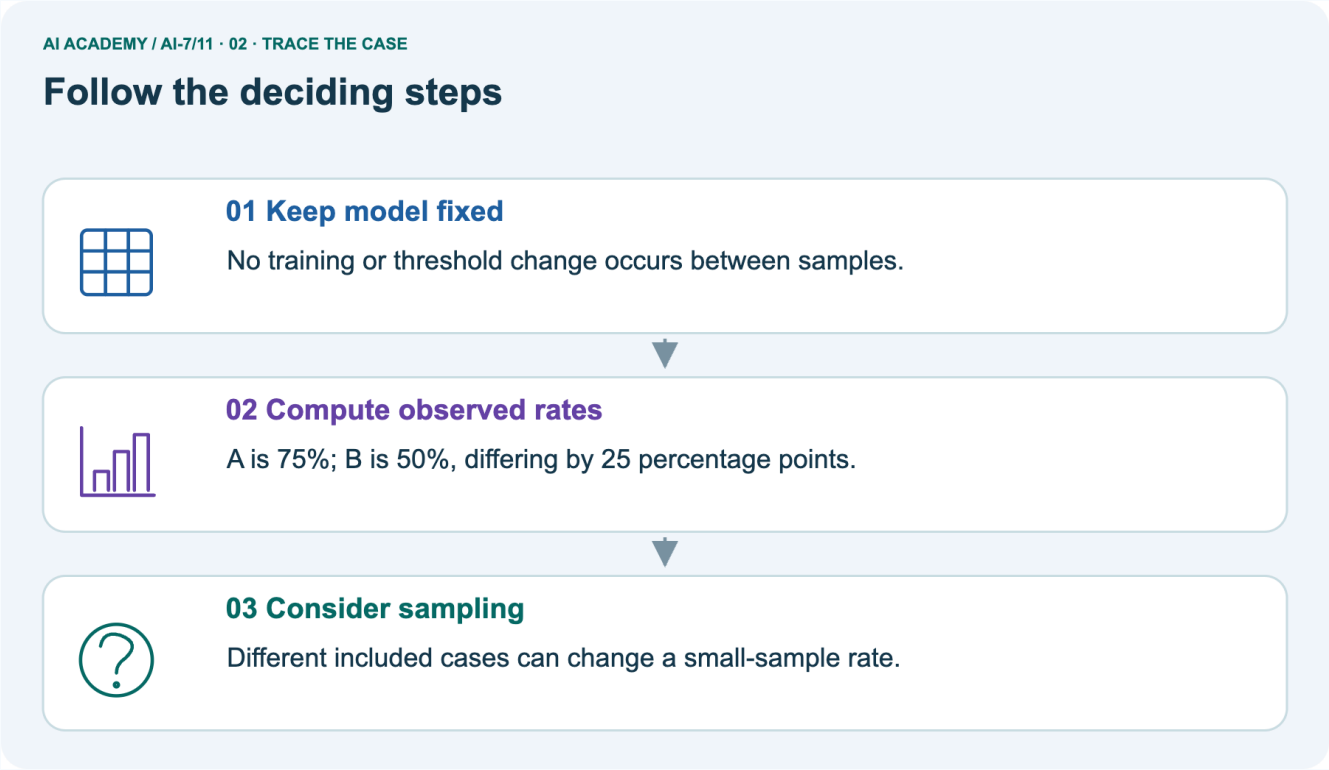
Follow the labelled picture with the matching explanation. Open the vector diagram

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection. Supplied facts: Sample A has 3 correct of 4; sample B has 2 correct of 4.

The model and rubric are unchanged.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

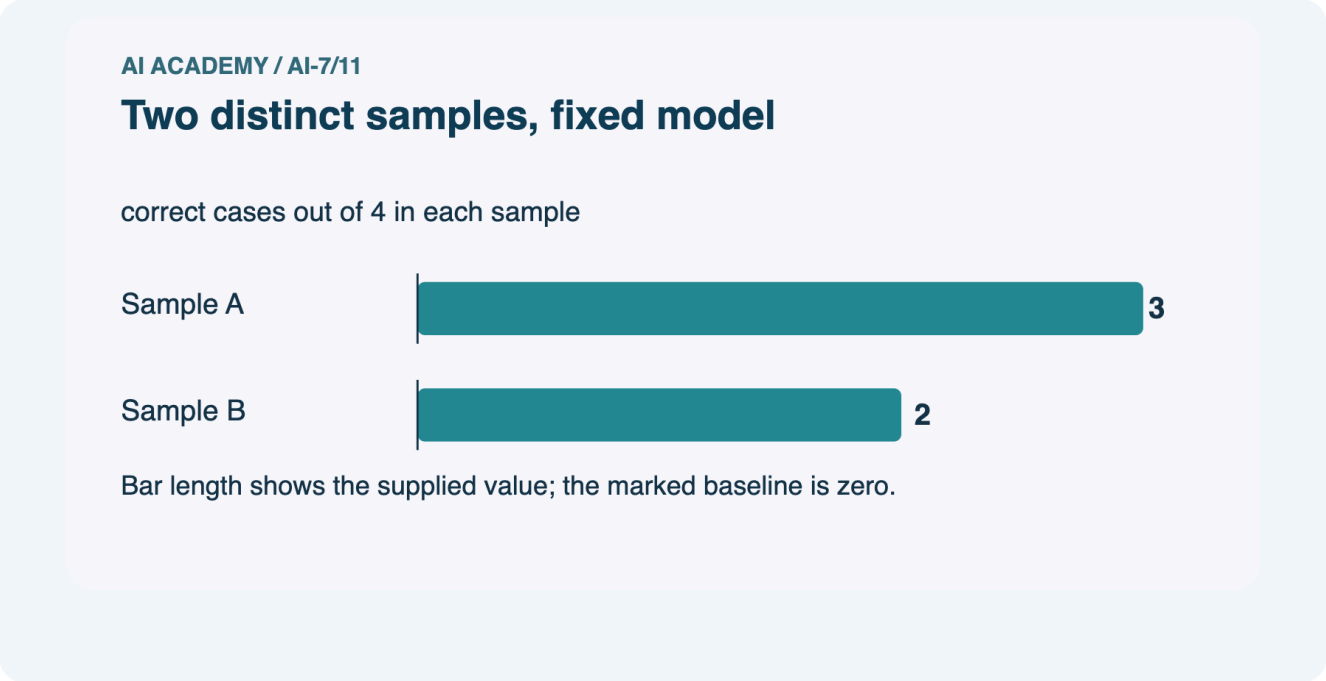


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep model fixed: No **training** or threshold change occurs between samples. → Compute observed rates: A is 75%; B is 50%, differing by 25 percentage points. → Consider sampling: Different included cases can change a small-sample rate.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two distinct samples, fixed model

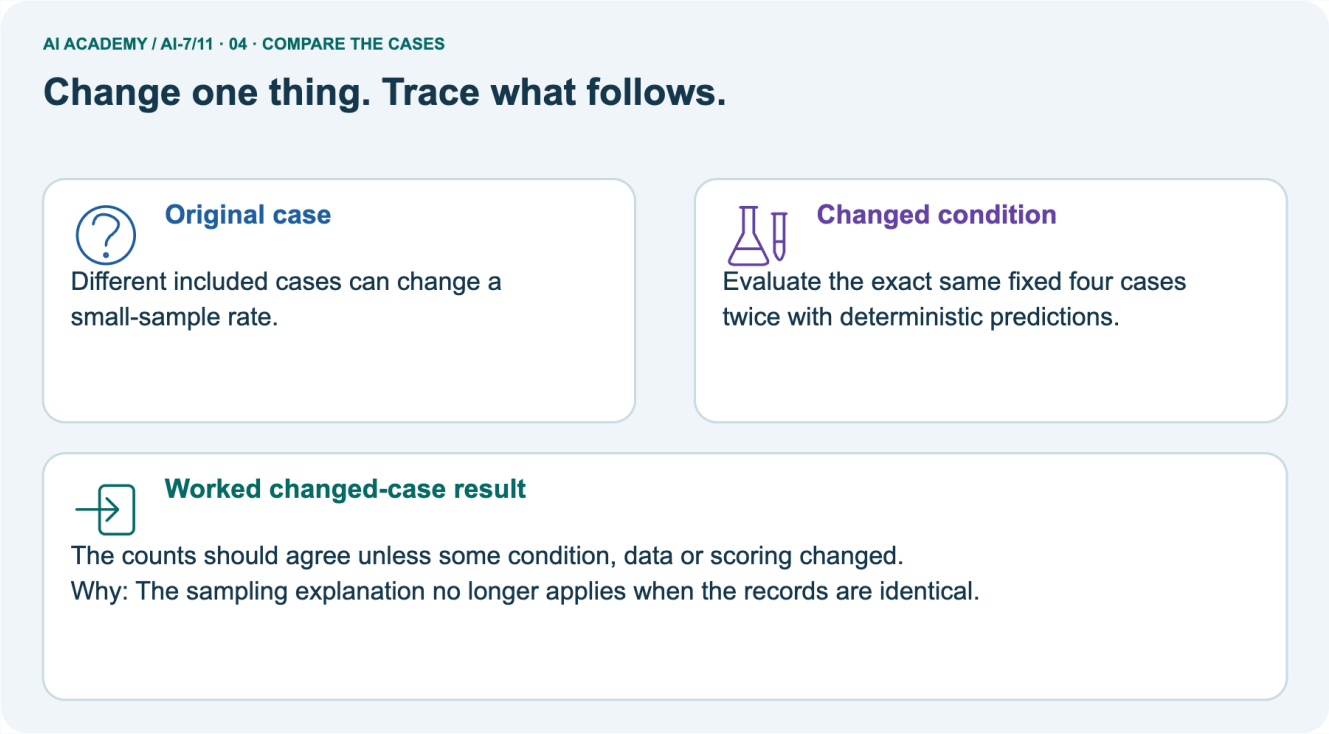


Follow the labelled picture with the matching explanation. Open the vector diagram

Two distinct samples, fixed model. Units: correct cases out of 4 in each sample. Sample A: 3. Sample B: 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



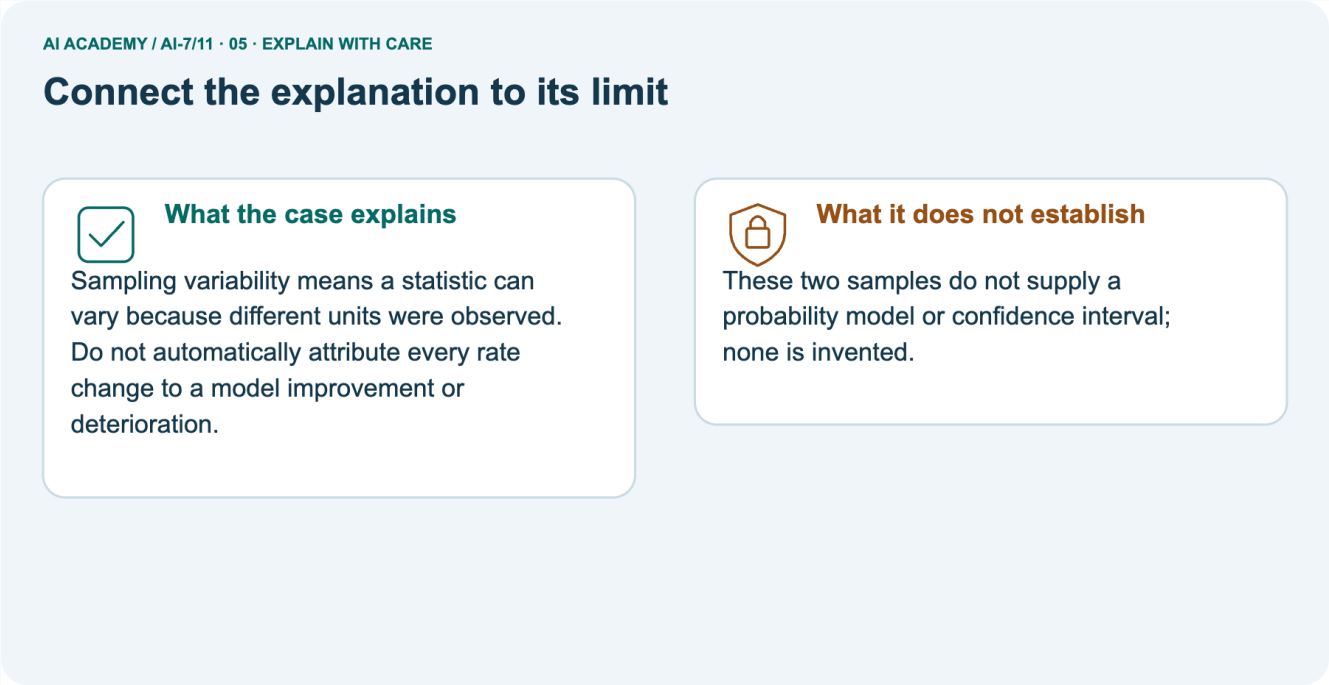
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Different included cases can change a small-sample rate. Changed condition: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

Expected result: The counts should agree unless some condition, data or scoring changed. Reason: The sampling explanation no longer applies when the records are identical.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



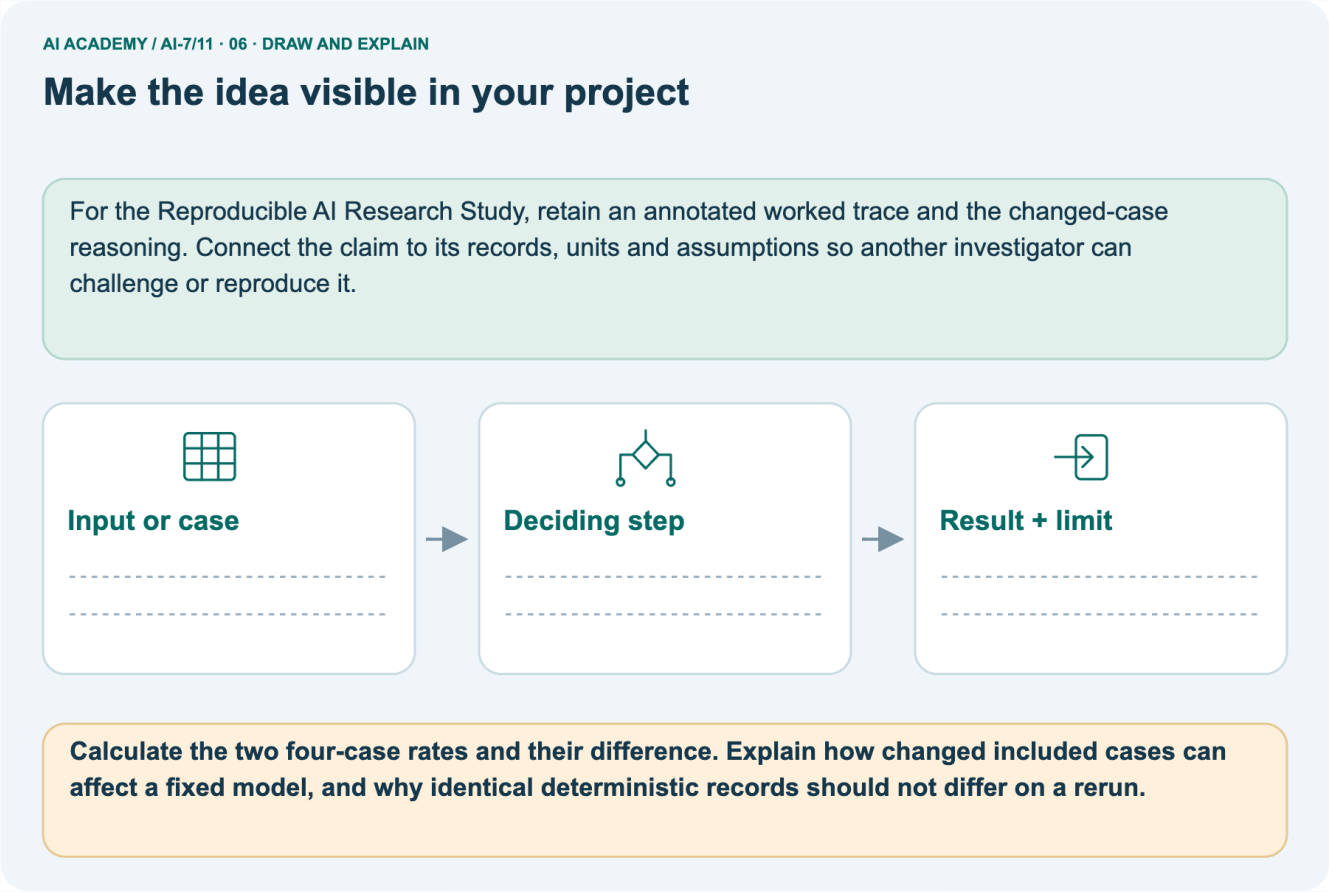
Follow the labelled picture with the matching explanation. Open the vector diagram

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Limit: These two samples do not supply a probability model or confidence interval; none is invented.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.